RNN-AGT: A Recurrent Neural Networks-Based Accelerated Gap-Time Model Using Gehan-Type Rank Loss

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Abstract

Recurrent event data, in which subjects experience the same type of event multiple times, frequently arise in medicine, insurance, reliability, and economics. Unlike the Cox proportional hazards model, which assumes covariates act multiplicatively on the hazard function, the accelerated failure time (AFT) model specifies a direct multiplicative effect on the event time scale, enabling prediction of survival times even when the proportional hazards assumption is violated. However, conventional AFT models rely on linear predictors that may be too restrictive when covariate effects are nonlinear or involve complex interactions. We propose RNN-AGT, a neural network–based survival model that extends the AFT framework to predict recurrent gap times. The model uses Gated Recurrent Units (GRUs) to capture temporal dependence while mitigating the vanishing gradient problem and maintaining lower computational cost than Long Short-Term Memory (LSTM) networks. To accommodate incomplete gap times, training is based on a Gehan-type rank loss with a subsampling method that reduces the computational burden of pairwise comparisons while preserving unbiased gradient estimates. Simulation studies show that RNN-AGT achieves strong predictive accuracy, measured by adjusted mean squared error (AMSE) and the inverse probability of censoring weighted (IPCW) concordance index, while capturing nonlinear and temporal patterns in repeated events. The method is further evaluated on two biomedical recurrent-event datasets and benchmarked against state-of-the-art Cox-based models.

Keywords: Recurrent event data, Accelerated gap-time model, Neural networks, Incomplete data, Gated Recurrent Units, Gehan-type rank loss

1 Introduction

Recurrent events arise when the same type of outcome can occur repeatedly for a given subject or unit over time. Such data are common in a wide range of domains, including repeated hospital admissions in chronic disease management, multiple insurance claims for a policyholder, successive failures of industrial equipment, and recurring economic disruptions. Modeling recurrent events accurately is essential for predicting future occurrences, evaluating

interventions, and supporting decisions about monitoring, prevention, and resource allocation.

Despite their importance, recurrent-event data remain challenging to analyze, especially in modern settings where covariate effects may be nonlinear, high-dimensional, and time-dependent. Classical hazard-based approaches provide a principled and interpretable framework. The Cox proportional hazards model (Cox, 1972) and several of its recurrent-event extensions, including the Prentice–Williams–Peterson (PWP) conditional models (Prentice et al., 1981), the Andersen–Gill (AG) counting-process model (Andersen and Gill, 1982), and the Wei–Lin–Weissfeld (WLW) marginal model (Wei et al., 1989), are widely used. However, their performance can deteriorate when the proportional hazards assumption is violated, when the linear predictor cannot capture complex covariate relationships, or when the data exhibit strong temporal dependence across repeated events. These difficulties are amplified in modern applications with rich longitudinal features and complex event dynamics (Zhang and Shi, 2024).

Accelerated failure time (AFT) models offer an alternative perspective by modeling event times directly as a function of covariates (Reid, 1994). However, standard AFT formulations typically rely on linear covariate effects, limiting their ability to represent interactions and nonlinearities. Recent advances in deep learning have therefore motivated a growing literature on neural network–based survival models. For single-event data, DeepSurv (Katzman et al., 2018) replaces the Cox linear predictor with a neural network to capture nonlinear covariate effects, while DeepAFT (Norman et al., 2024) extends AFT modeling via losses based on the Buckley–James estimator or inverse-probability-of-censoring weighting (IPCW), though IPCW can be unstable in finite samples. Rank-based deep AFT methods, such as DeepR-AFT (Kim et al., 2024), leverage Gehan-type weighting (Gehan, 1965) and have demonstrated competitive predictive performance against random survival forests and classical parametric and semiparametric AFT approaches. Nevertheless, most deep survival methods remain tailored to single-event outcomes, and their direct extension to recurrent gap-time settings is not straightforward. In particular, recurrent gap times cannot be treated as naive clustered observations without risking model misspecification and invalid inference (Luo and Huang, 2010).

This paper addresses these gaps by introducing RNN-AGT, a recurrent neural network framework for recurrent gap-time data that combines the AFT paradigm with sequential modeling. RNN-AGT uses Gated Recurrent Units (GRUs) introduced by Cho et al. (2014), to learn temporal dependence across successive event intervals, providing a practical compromise between interpretability and computational efficiency. Compared with Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997), GRUs typically require fewer parameters while still mitigating vanishing-gradient issues that affect standard recurrent neural networks. Recurrent neural architectures have been explored primarily within proportional-hazards formulations, leading to models such as RNN-SURV (Giunchiglia et al., 2018), Deep Recurrent Survival Analysis (DRSA) (Ren et al., 2019), and LSTM-Cox (Zhang and Shi, 2024). In contrast, RNN-AGT is designed to operate directly within an AFT gap-time framework.

To accommodate censoring and incomplete observation times, features that are central not only in single-event survival analysis (Kaplan and Meier, 1958; Ritov, 1990) but also in many applied settings (Klein and Moeschberger, 2006; Wang et al., 2017) including recurrent-

event applications (Lin et al., 1998; Strawderman, 2005, 2006; Lyu et al., 2018), RNN-AGT is trained using a Gehan-type rank-based loss. Because rank-based objectives can be computationally demanding for large samples, we further incorporate a subsampling strategy to reduce computational complexity, building on ideas developed in the subsampling literature (Ma et al., 2014; Wang et al., 2018; Wang and Ma, 2021) and recently used in deep AFT training (Kim et al., 2024). We evaluate the proposed framework through extensive simulation studies and applications to biomedical datasets. Across these experiments, RNN-AGT demonstrates a strong ability to capture nonlinear covariate effects and temporal patterns in recurrent gap times, while achieving predictive performance that matches state-of-the-art Cox-based recurrent-event approaches.

The remainder of the paper is structured as follows: In Section 2, we introduce the RNN-AGT framework, describe the rank-based objective and the subsampling strategy that makes training scalable, and discuss practical modeling choices. Section 3 then examines the finite-sample behavior of the proposed approach through extensive simulation studies. In Section 4, we turn to real biomedical datasets to illustrate how RNN-AGT captures temporal dependence and nonlinear covariate effects in practice, and to benchmark its predictive performance against established recurrent-event models. Section 5 concludes by summarizing the main contributions, noting limitations, and outlining promising directions for future research.

2 Methodology

Let $i = 1, \ldots, n$ index subjects (or units) and let $j = 1, 2, \ldots$ index recurrent events for subject i. Denote by S_{ij} the calendar time of the jth event for subject i, with $S_{i0} = 0$, and define the corresponding gap time by $T_{ij} = S_{ij} - S_{i,j-1}$. Our goal is to model and predict the sequence $\{T_{ij}\}_{j\geq 1}$ in settings where covariate effects may be nonlinear and where successive events may exhibit temporal dependence, while also accommodating incomplete gap times.

RNN-AGT (Recurrent Neural Network for Accelerated Gap Times) is built on an accelerated failure time (AFT) formulation for gap times together with a recurrent representation of event history. Rather than imposing a linear predictor, RNN-AGT models the conditional mean of $\log T_{ij}$ through a flexible nonlinear function learned from the observed recurrent-event trajectories.

2.1 AFT Formulations for Recurrent Gap Times

2.1.1 Recurrent Gap Time Data

To construct the rank-based loss, we first describe the recurrent gap-time data structure following Lyu et al. (2018). In practice, subject observation may end due to termination of monitoring. Let C_i denote the monitoring time for subject i, and assume that C_i is independent of the recurrent gap-time process conditional on the covariates $\mathbf{Z}_i$. We then define K_i^* as the number of gap times for subject i that are fully observed before the monitoring time.

$$K_i = \max\left\{k \geq 0 : \sum_{j=1}^{k} T_{ij} \leq C_i\right\},$$

so that

$$\sum_{j=1}^{K_i} T_{ij} \leq C_i < \sum_{j=1}^{K_i+1} T_{ij}.$$

For each $j \geq 1$, define the event indicator

$$\delta_{ij} = \mathbb{I}\left(\sum_{l=1}^{j} T_{il} \leq C_i\right),$$

so that $\delta_{ij} = 1$ if the jth gap time is fully observed and $\delta_{ij} = 0$ otherwise.

The observed time contribution at index j is defined by

$$G_{ij} = \min\left(T_{ij},\ C_i - \sum_{l=1}^{j-1} T_{il}\right),$$

which equals the true gap time when $\delta_{ij} = 1$ and equals the remaining monitoring time when the gap time is incompletely observed. For notational convenience, we also write $Y_{ij} = \log(G_{ij})$.

Hence, the observed data for subject i can be expressed as

$$\mathcal{O}_i = \Big\{\big(G_{ij}, \delta_{ij}\big) : j = 1, \ldots, K_i + 1;\ \mathbf{Z}_i,\ C_i\Big\}.$$

2.1.2 Linear AFT Specification

Let $\mathbf{Z}_i \in \mathbb{R}^p$ denote the baseline covariate vector for subject i. A classical linear AFT specification for recurrent gap times posits that the observed gap times satisfy

$$Y_{ij} = \boldsymbol{\theta}_0' \mathbf{Z}_i + \varepsilon_{ij}, \qquad j \geq 1, \tag{2.1}$$

where $\boldsymbol{\theta}_0 \in \mathbb{R}^p$ is the true regression parameter and ε_{ij} captures unobserved heterogeneity. For each subject i, write $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \ldots)^\top$. We assume that the errors $\{\varepsilon_{ij}\}_{j\geq 1}$ share a common marginal distribution across event indices j for a given subject, while placing no parametric restrictions on their within-subject dependence. This is less restrictive than a shared frailty framework, which introduces a latent subject effect and often assumes conditional independence of gap times given that effect. By leaving the dependence structure unspecified, (2.1) can accommodate a broad range of correlation patterns across repeated gap times, including autoregressive and other forms of temporal dependence (Lyu et al., 2018).

2.1.3 Nonlinear AFT Specification

On the log-gap scale, RNN-AGT assumes

$$Y_{ij} = f_{\mathbf{w}}\big(\mathbf{h}_{i,j-1}, \mathbf{x}_{ij}\big) + \varepsilon_{ij},$$

where $f_{\boldsymbol{\theta}}(\cdot)$ is a nonlinear mapping parameterized by the network weights $\mathbf{w}$. The hidden state $\mathbf{h}_{ij} \in \mathbb{R}^d$ is intended to summarize the relevant information about subject i's history up to the jth event, and $\mathbf{x}_{ij}$ denotes the input provided to the recurrent unit at step j. The term ε_{ij} captures unexplained variation. In contrast to parametric AFT models, we do not specify a distribution for ε_{ij}. Hence, given the predicted gap times, event times are recovered by the recursion $S_{ij} = S_{i,j-1} + G_{ij}$, preserving the natural relationship between gap times and the event process.

2.1.4 Hidden State and the Role of Covariates

The hidden state $\mathbf{h}_{ij}$ provides the mechanism by which the model adapts predictions as events accumulate and the event-generating mechanism evolves.

Two natural ways to incorporate baseline covariates are possible. In the first, baseline covariates are concatenated to the sequential input at every time step. In the second, baseline covariates are encoded once into the initial hidden state, and the recurrent updates focus on the evolution of the gap-time sequence itself. Throughout this work we adopt the second approach, which is particularly attractive when the event history is long or the covariate dimension is large.

Specifically, baseline covariates are incorporated through the initialization of the hidden state. Let $\mathbf{Z}_i \in \mathbb{R}^p$ denote the vector of time-fixed baseline covariates for subject i. We initialize the hidden state as

$$\mathbf{h}_{i0} = \tanh\big(W_{\text{init}}\mathbf{Z}_i + \mathbf{b}_{\text{init}}\big), \tag{2.2}$$

where $\tanh(\cdot)$ is the hyperbolic tangent function, W_{init} is a trainable weight matrix mapping the baseline covariates to the hidden-state space, and $\mathbf{b}_{\text{init}}$ is a bias vector. This initialization allows baseline information to influence the entire subsequent trajectory of the latent representation. The recurrent input at event index j is taken to be the previous log-gap time,

$$\mathbf{x}_{ij} = Y_{i,j-1}, \tag{2.3}$$

and the network maps the updated hidden state to a prediction of the conditional mean

$$\widehat{\mu}_{ij} = \mathbf{w}^\top \mathbf{h}_{ij} + b, \tag{2.4}$$

where $\mathbf{w}$ is a trainable weight vector and b is a scalar bias term defining the linear output layer. The quantity $\widehat{\mu}_{ij}$ serves as the model-based estimate of $\mathbb{E}(Y_{ij} \mid \mathcal{F}_{i,j-1})$, with $\mathcal{F}_{i,j-1}$ denoting the information available up to event $j-1$ for subject i.

Encoding baseline covariates in the initial state yields a separation between static and dynamic components. Baseline characteristics act through $\mathbf{h}_{i0}$, while the recurrent evolution of $\mathbf{h}_{ij}$ is driven by the event history. This reduces redundant processing of static features

across time, improves computational efficiency, and tends to regularize the model by limiting repeated exposure to the same covariate vector at each step.

The formulation also admits a straightforward extension to time-varying covariates. If additional covariates $\mathbf{Z}_{ij}$ are observed at each event, they can be appended to the sequential input by redefining $\mathbf{x}_{ij}$ to include both $Y_{i,j-1}$ and $\mathbf{Z}_{ij}$, while baseline covariates remain encoded in $\mathbf{h}_{i0}$.

2.2 GRU Architecture for Sequential Dependence

Training recurrent neural networks on long event histories can be challenging because the learning signal must be propagated backward through many time steps. Under backpropagation through time, gradients of the loss with respect to early parameters are formed by repeated applications of the chain rule, yielding products of Jacobian terms across the sequence. As a consequence, gradient magnitudes can become highly unbalanced across time: when the factors in these products are typically smaller than one, a common situation with bounded nonlinearities such as tanh, gradients reaching early time steps can decay rapidly, producing the well-known *vanishing gradient* problem. In the opposite regime, these products may grow without bound, yielding *exploding gradients* and unstable optimization (Rehmer and Kroll, 2020). Both phenomena can slow convergence and make it difficult for standard RNNs to learn long-range temporal dependence, which is essential in recurrent-event modeling where earlier events may influence later gap times.

These limitations motivate architectures that explicitly control information flow across time. In RNN-AGT, we therefore model temporal dependence using a Gated Recurrent Unit (Cho et al., 2014). GRUs introduce gating mechanisms that adaptively combine new inputs with the existing hidden state, which can improve optimization behavior and facilitate learning from long sequences. Moreover, compared with Long Short-Term Memory networks (Hochreiter and Schmidhuber, 1997), GRUs often achieve similar expressive power with fewer parameters, which is attractive in recurrent-event applications where sequence lengths vary and sample sizes may be moderate. Complementary to architectural choices, additional optimization techniques have been proposed to improve gradient propagation in deep networks, such as gradient amplification (Basodi et al., 2020).

Let $\mathbf{h}_{ij}$ denote the hidden state after processing the $(j-1)$st gap time. With input $\mathbf{x}_{ij}$, the GRU update is given by

$$\mathbf{z}_{ij} = \sigma(W_z \mathbf{x}_{ij} + U_z \mathbf{h}_{i,j-1} + \mathbf{b}_z)\,, \tag{2.5}$$

$$\mathbf{r}_{ij} = \sigma(W_r \mathbf{x}_{ij} + U_r \mathbf{h}_{i,j-1} + \mathbf{b}_r)\,, \tag{2.6}$$

$$\tilde{\mathbf{h}}_{ij} = \tanh(W_h \mathbf{x}_{ij} + U_h\,(\mathbf{r}_{ij} \odot \mathbf{h}_{i,j-1}) + \mathbf{b}_h)\,, \tag{2.7}$$

$$\mathbf{h}_{ij} = (1 - \mathbf{z}_{ij}) \odot \mathbf{h}_{i,j-1} + \mathbf{z}_{ij} \odot \tilde{\mathbf{h}}_{ij}, \tag{2.8}$$

where $\sigma(\cdot)$ denotes the logistic function and $\odot$ denotes element-wise multiplication. The matrices W_z, W_r, W_h represent input weight parameters, U_z, U_r, U_h are recurrent weight matrices associated with the previous hidden state, and $\mathbf{b}_z, \mathbf{b}_r, \mathbf{b}_h$ are bias vectors. The update gate $\mathbf{z}_{ij}$ controls how strongly the hidden state is refreshed at each step, while the reset gate $\mathbf{r}_{ij}$ determines the extent to which past information contributes to the candidate state $\tilde{\mathbf{h}}_{ij}$. The predicted mean $\widehat{\mu}_{ij}$ is then obtained from $\mathbf{h}_{ij}$ through (2.4).

Deeper specifications can be obtained by stacking multiple GRU layers, and dropout may be applied between layers to reduce overfitting in long sequences or high-dimensional applications. These architectural variations do not change the underlying modeling principle: RNN-AGT remains a nonparametric AFT model for recurrent gap times in which temporal dependence is learned through a recurrent hidden representation.

The next subsection introduces the rank-based loss used for incomplete gap times and the subsampling method used to reduce computational complexity.

2.3 Gehan-Type Rank-Based Loss Function

This subsection constructs the objective function used to train RNN-AGT under incomplete monitoring. The development follows the rank-based AFT approach for censored data, where estimation can proceed without specifying the error distribution. Such semiparametric AFT ideas trace back to classical censored regression work (e.g., Buckley and James (1979)) and to Gehan-type weighting schemes for right-censored data (Gehan, 1965). We adapt these ideas to recurrent gap times using the weighted risk-set (WRS) methods of Luo and Huang (2010), which effectively account for the sequential nature of recurrent gap time data.

2.3.1 Residual scale and risk-set representation

Let G_{ij} denote the jth gap time for subject i, and let $\delta_{ij} \in \{0, 1\}$ indicate whether G_{ij} is observed ($\delta_{ij} = 1$) or incomplete due to end of monitoring ($\delta_{ij} = 0$). Let $\mathbf{Z}_i \in \mathbb{R}^p$ be the baseline covariate vector for subject i.

For a linear AFT model, define the residual on the log-gap scale by

$$e_{ij}(\boldsymbol{\theta}) = Y_{ij} - \boldsymbol{\theta}'\mathbf{Z}_i, \tag{2.9}$$

where $\boldsymbol{\theta} \in \mathbb{R}^p$ is an unknown regression parameter. Following standard censored-data constructions, define the residual-scale counting and at-risk processes

$$N_{ij}(\boldsymbol{\theta}, t) = \delta_{ij}\, \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \le t\}, \qquad R_{ij}(\boldsymbol{\theta}, t) = \mathbb{I}\{e_{ij}(\boldsymbol{\theta}) \ge t\}. \tag{2.10}$$

2.3.2 WRS method for recurrent gap times

A naive rank formulation that pools all (i, j) pairs would implicitly upweight subjects with long observed event histories. To ensure that each subject contributes comparably, let K_i be the number of observed events for subject i and define

$$K_i^* = \max\{K_i, 1\}.$$

Each subject contributes K_i^* retained gap-time records, and each record receives weight $1/K_i^*$. This weighting yields subject-level averaged processes

$$N_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} N_{ij}(\boldsymbol{\theta}, t), \qquad R_i^*(\boldsymbol{\theta}, t) = \frac{1}{K_i^*} \sum_{j=1}^{K_i^*} R_{ij}(\boldsymbol{\theta}, t), \tag{2.11}$$

and the empirical risk-set summaries

$$S_0^*(\boldsymbol{\theta}, t) = \frac{1}{n}\sum_{i=1}^{n} R_i^*(\boldsymbol{\beta}, t), \qquad S_1^*(\boldsymbol{\theta}, t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{Z}_i\, R_i^*(\boldsymbol{\theta}, t). \tag{2.12}$$

Related rank-based developments for clustered and correlated censored outcomes further support the use of such weighting-aggregation tools when repeated times arise within subjects (Wang and Fu, 2011).

2.3.3 Gehan-type weighting and the induced loss

A broad class of weighted risk-set (WRS) rank estimating functions can be written in counting-process form as

$$U(\boldsymbol{\theta}) = \sum_{i=1}^{n} \int W^*(t \mid \boldsymbol{\theta}) \left\{ \mathbf{Z}_i - \frac{S_1^*(\boldsymbol{\theta}, t)}{S_0^*(\boldsymbol{\theta}, t)} \right\} dN_i^*(\boldsymbol{\theta}, t), \tag{2.13}$$

where $W^*(t \mid \boldsymbol{\theta})$ is a weight function. A Gehan-type choice specifies

$$W^*(t \mid \boldsymbol{\theta}) = S_0^*(\boldsymbol{\theta}, t), \tag{2.14}$$

which mirrors the classical Gehan weighting in censored rank procedures (Gehan, 1965). Under this choice, the estimating equation admits an equivalent pairwise-comparison interpretation: the criterion aggregates comparisons of residuals within risk sets, so that estimation is driven by the ordering of residuals rather than their absolute scale (Luo and Huang, 2010).

As in the single-event AFT literature, the Gehan-type estimating function can be viewed as the gradient of a convex rank-based objective function. In the WRS recurrent gap-time setting, this is derived as

$$L_{\mathrm{G}}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K_i^*}\sum_{l=1}^{n}\sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\boldsymbol{\theta}) - e_{ij}(\boldsymbol{\theta}) \Big]^-, \tag{2.15}$$

where $e_{ij}(\boldsymbol{\theta})$ denotes the residual for the jth gap time of subject i as defined in (2.9), δ_{ij} is the corresponding event indicator, and $[a]^- = |\min(a, 0)|$ denotes the negative part. The loss in (2.15) generalizes the classical Gehan loss to recurrent gap times by summing, with WRS normalization, the extent to which other residuals fall below each completely observed residual. In particular, δ_{ij} ensures that only fully observed gap times contribute to the pairwise comparisons.

Related rank-based loss constructions have been studied for censored data. For example, Strawderman (2005) proposed a class of weighted log-rank-type loss functions motivated by semiparametric efficiency theory (Andersen et al., 1993; van der Vaart, 1998). Such objectives can be expressed through stochastic-integral representations closely related to (2.13); see also Peña et al. (2001) for related derivations on the baseline gap-time scale.

Although $L_{\mathrm{G}}(\boldsymbol{\theta})$ is convex, it is non-smooth. Closed-form solutions for the corresponding Gehan-type estimator, denoted here by $\widehat{\boldsymbol{\theta}}_G$, are therefore not available in general. A standard approach is to compute $\widehat{\boldsymbol{\theta}}_G$ by minimizing (2.15) using linear programming techniques,

following the approach used for rank-based AFT estimation in Lin et al. (1998). In addition, direct optimization can be numerically challenging due to the non-differentiability of the negative-part function. To improve numerical stability and facilitate variance estimation, induced smoothing can be used, which replaces the non-smooth estimating equation (or its objective) with a smooth approximation while retaining the same target parameter (Lyu et al., 2018). Finally, note that minimizing $L_{\mathrm{G}}(\boldsymbol{\theta})$ is equivalent to solving the Gehan-type estimating equation in the sense that $U(\boldsymbol{\theta})$ is proportional to the gradient of $L_{\mathrm{G}}(\boldsymbol{\theta})$, so their solutions coincide under standard regularity conditions (Fygenson and Ritov, 1994).

2.3.4 Nonlinear Extension

We now extend the Gehan-type rank-based loss function in (2.15), which is derived under a linear predictor, to a nonlinear predictor while remaining within AFT paradigm. Specifically, consider the recurrent gap-time data $\boldsymbol{\mathcal{O}} = \{(G_{ij}, \delta_{ij}, \mathbf{Z}_i, C_i) : i = 1, \ldots, n,\ j = 1, \ldots, K_i^*\}$ obtained after the WRS transformation, with $K_i^* = \max(K_i, 1)$ where K_i the number of observed events for subject i.

Theorem 2.1. *Let $\widehat{\mu}_{ij}(\mathbf{w})$ denote the history-dependent nonlinear predictor produced by RNN-AGT, with trainable parameters $\mathbf{w} \in \mathcal{W}$. The induced nonlinear residual on the log-gap-time scale is*

$$e_{ij}(\mathbf{w}) = Y_{ij} - \widehat{\mu}_{ij}(\mathbf{w}),$$

and the Gehan-type rank-based WRS training loss for the nonlinear accelerated gap-time model is

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-}. \tag{2.16}$$

If the nonlinear predictor is restricted to the linear $\widehat{\mu}_{ij}(\mathbf{w}) = \boldsymbol{\theta}'\mathbf{Z}_i$, then (2.16) reduces to the corresponding WRS Gehan objective for recurrent gap times obtained by Lyu et al. (2018). In the single-event case ($K_i^* \equiv 1$), the nonlinear loss becomes the Gehan-type rank objective used in deep AFT modeling by Kim et al. (2024).

Proof. The Gehan-type WRS objective in (2.15) depends on $\boldsymbol{\theta}$ only through the residuals $\{e_{ij}(\boldsymbol{\theta})\}$, where $e_{ij}(\boldsymbol{\theta}) = Y_{ij} - \boldsymbol{\theta}'\mathbf{Z}_i$. Let $\widehat{\mu}_{ij}(\mathbf{w})$ be any (possibly history-dependent) nonlinear predictor and define the induced AFT residual $e_{ij}(\mathbf{w}) = Y_{ij} - \widehat{\mu}_{ij}(\mathbf{w})$. Replacing $e_{ij}(\boldsymbol{\theta})$ by $e_{ij}(\mathbf{w})$ in (2.15) yields

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \sum_{l=1}^{n} \sum_{k=1}^{K_l^*} \frac{\delta_{ij}}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-},$$

which is (2.16). If $\widehat{\mu}_{ij}(\mathbf{w}) = \boldsymbol{\theta}'\mathbf{Z}_i$, then $e_{ij}(\mathbf{w}) = e_{ij}(\boldsymbol{\theta})$ and the loss reduces to (2.15). If $K_i^* \equiv 1$, the WRS weights are identically one and the loss reduces to the single-event Gehan-type rank objective.

Moreover, while $L_{\mathrm{G}}(\boldsymbol{\theta})$ is convex in the parameter $\boldsymbol{\theta}$, the nonlinear loss $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ is generally not convex in $\mathbf{w}$ because the map $\mathbf{w} \mapsto \widehat{\mu}_{ij}(\mathbf{w})$ induced by a neural network is nonlinear. In addition, the dimension of $\mathbf{w}$ can be much larger than p and increases with the number of layers, hidden units, and recurrent components, which makes optimization of $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ substantially more computationally demanding than its linear counterpart. □

2.4 Sub-sampling Pairs

The nonlinear Gehan-type objective $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ in (2.16) is defined through pairwise comparisons of residuals across all retained gap times. Let $N^* = \sum_{i=1}^{n} K_i^*$ denote the total number of retained gap-time records after the WRS transformation. A direct evaluation of $L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w})$ requires $O((N^*)^2)$ pairwise terms, which quickly becomes prohibitive when subjects experience too many recurrent events.

To obtain a more efficient training procedure, we adopt a pair sub-sampling method that enables mini-batch stochastic optimization while preserving unbiasedness at the level of the loss (under suitable regularity conditions).

Let $\mathcal{I} = \{(i,j) : i = 1, \ldots, n,\ j = 1, \ldots, K_i^*\}$ be the index set of all retained records and let

$$\mathcal{U} = \{(i,j) \in \mathcal{I} : \delta_{ij} = 1\}$$

denote the subset of fully observed gap times, with cardinality $n' = |\mathcal{U}| = \sum_{i=1}^{n} \sum_{j=1}^{K_i^*} \delta_{ij}$. For each pair $(i,j) \in \mathcal{U}$, we draw s indices independently (with replacement) from $\mathcal{I}$, denoted by

$$(l,k) \sim q_{ij}(\cdot) \quad \text{on } \mathcal{I},$$

where q_{ij} is a user-specified sampling distribution (for example, uniform over $\mathcal{I}$, or uniform over $\mathcal{I} \setminus \{(i,j)\}$). This produces a multiset of sampled ordered pairs

$$\mathcal{P}_s = \Big\{ \big((i,j),(l,k)\big) : (i,j) \in \mathcal{U},\ (l,k) \text{ is drawn } s \text{ times from } q_{ij} \Big\},$$

with total size $m = n's$. When q_{ij} is uniform over $\mathcal{I}$, the resulting cost per epoch is $O(n's)$.

The full nonlinear Gehan objective can be written as

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \sum_{(l,k)\in\mathcal{I}} \frac{1}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-}.$$

The following result gives a mini-batch objective whose expectation equals the full loss.

Theorem 2.2. *Fix an integer $s \geq 1$. For each $(i,j) \in \mathcal{U}$, let $(l,k)_{ij,1}, \ldots, (l,k)_{ij,s}$ be i.i.d. draws from a sampling distribution q_{ij} on $\mathcal{I}$ such that $q_{ij}(l,k) > 0$ for all $(l,k) \in \mathcal{I}$. Define the sub-sampled loss*

$$\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \frac{1}{s} \sum_{r=1}^{s} \frac{1}{K_i^* K_{l_r}^*} \frac{1}{q_{ij}\big((l,k)_{ij,r}\big)} \Big[e_{(l,k)_{ij,r}}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^{-}, \tag{2.17}$$

where $K_{l_r}^*$ denotes K_l^* for the sampled index $(l,k)_{ij,r}$. Then, for every $\mathbf{w} \in \mathcal{W}$,

$$\mathbb{E}\Big[\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w}) \,\Big|\, \boldsymbol{\mathcal{O}} \Big] = L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}),$$

where the expectation is taken over the random draws $\{(l,k)_{ij,r}\}$. In particular, when q_{ij} is uniform on $\mathcal{I}$, i.e., $q_{ij}(l,k) = 1/N^*$, the importance weight $1/q_{ij}$ is constant and (2.17) simplifies to a scaled average of $m = n's$ sampled pair contributions.

Proof. Fix $\mathbf{w} \in \mathcal{W}$. Write the full loss as

$$L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}), \qquad a_{ij}(l,k;\mathbf{w}) = \frac{1}{K_i^* K_l^*} \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^-.$$

For each fixed $(i,j) \in \mathcal{U}$, let $(l,k)_{ij,1}, \ldots, (l,k)_{ij,s}$ be i.i.d. from q_{ij} on $\mathcal{I}$. Define the single-draw importance-weighted contribution

$$X_{ij,r}(\mathbf{w}) = \frac{1}{K_i^* K_{l_r}^*} \frac{1}{q_{ij}\big((l,k)_{ij,r}\big)} \Big[e_{(l,k)_{ij,r}}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^- = \frac{a_{ij}\big((l,k)_{ij,r};\mathbf{w}\big)}{q_{ij}\big((l,k)_{ij,r}\big)}.$$

Since $q_{ij}(l,k) > 0$ for all $(l,k) \in \mathcal{I}$, we have

$$\mathbb{E}[X_{ij,r}(\mathbf{w}) \mid \boldsymbol{\mathcal{O}}] = \sum_{(l,k)\in\mathcal{I}} q_{ij}(l,k) \frac{a_{ij}(l,k;\mathbf{w})}{q_{ij}(l,k)} = \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}).$$

By i.i.d. sampling, $\mathbb{E}\big[\frac{1}{s}\sum_{r=1}^{s} X_{ij,r}(\mathbf{w}) \mid \boldsymbol{\mathcal{O}}\big] = \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w})$. Therefore,

$$\mathbb{E}\Big[\widetilde{L}_{\mathrm{G},s}^{\mathrm{NL}}(\mathbf{w}) \,\Big|\, \boldsymbol{\mathcal{O}}\Big] = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \mathbb{E}\left[\frac{1}{s} \sum_{r=1}^{s} X_{ij,r}(\mathbf{w}) \,\middle|\, \boldsymbol{\mathcal{O}} \right] = \frac{1}{n} \sum_{(i,j)\in\mathcal{U}} \sum_{(l,k)\in\mathcal{I}} a_{ij}(l,k;\mathbf{w}) = L_{\mathrm{G}}^{\mathrm{NL}}(\mathbf{w}).$$

Finally, if q_{ij} is uniform on $\mathcal{I}$, then $q_{ij}(l,k) = 1/N^*$ and the importance weight is constant, so (2.17) reduces (up to the constant factor N^*) to an average of $m = n's$ sampled pairwise terms. □

In practice, we form mini-batches by selecting a subset $\mathcal{B} \subset \mathcal{P}_s$ of size b from the sampled pairs and computing the corresponding mini-batch objective

$$\widetilde{L}_{\mathrm{G},\mathcal{B}}^{\mathrm{NL}}(\mathbf{w}) = \frac{1}{|\mathcal{B}|} \sum_{\big((i,j),(l,k)\big)\in\mathcal{B}} \frac{\delta_{ij}}{K_i^* K_l^*} \omega_{ij}(l,k) \Big[e_{lk}(\mathbf{w}) - e_{ij}(\mathbf{w}) \Big]^-, \tag{2.18}$$

where $\omega_{ij}(l,k) = 1/q_{ij}(l,k)$ is the importance weight (and can be dropped when q_{ij} is uniform, up to a constant scaling). Gradient-based optimizers are then applied to update $\mathbf{w}$ using $\nabla_{\mathbf{w}} \widetilde{L}_{\mathrm{G},\mathcal{B}}^{\mathrm{NL}}(\mathbf{w})$, yielding an efficient stochastic approximation to the full-batch optimization of (2.16).

3 Simulation Studies

3.1 Settings

To evaluate the finite-sample performance of the proposed RNN-AGT model for recurrent gap-time outcomes, we conducted extensive simulation studies in the spirit of Kim et al. (2024). Conditional on baseline covariates, log-gap times were generated according to

$$y_{ij} = f(\mathbf{z}_i) + \varepsilon_{ij}, \tag{3.1}$$

where $\mathbf{z}_i = (z_{i1}, z_{i2}, z_{i3})'$ is the baseline covariate vector, $f(\cdot)$ is the mean function, and ε_{ij} is an error term. Covariates were generated sequentially to induce dependence among predictors: $z_{i1} \sim \text{Bernoulli}(0.5)$, $z_{i2} \sim \mathcal{N}(z_{i1}/3, 1)$, and $z_{i3} \sim \mathcal{N}(z_{i2}/3, 1)$. We considered three specifications for $f(\cdot)$: an interaction-based nonlinear function $f_1(\mathbf{z}) = 3z_1 + 7z_1z_2 - 5z_3$, a GAM-type nonlinear function $f_2(\mathbf{z}) = z_1 + z_2^3 + \exp(0.9z_3)$, and a linear function $f_3(\mathbf{z}) = z_1 - 6z_2 + 4z_3$. Error terms were generated from three standardized distributions (standard normal, Gumbel, and logistic), each rescaled to have mean zero and variance one. To introduce incomplete gap times, a monitoring time C_i was generated as

$$C_i = \tau U, \qquad U \sim \text{Uniform}(0, 1), \tag{3.2}$$

with τ tuned to yield approximate proportions of 25%, 50%, and 65%. To incorporate within-subject dependence among repeated gap times, we examined two correlation mechanisms. First, we used a subject-specific random effect (frailty) $b_i \sim \mathcal{N}(0, \sigma_b^2)$ and set

$$\varepsilon_{ij} = b_i + \eta_{ij}, \qquad \eta_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{3.3}$$

which induces positive correlation among gap times within a subject. Second, we considered an autoregressive dependence structure

$$\eta_{ij} = \rho\, \eta_{i,j-1} + \sqrt{1-\rho^2}\, \xi_{ij}, \qquad \xi_{ij} \overset{i.i.d.}{\sim} \text{Error Distribution}, \tag{3.4}$$

with $\rho = 0.5$ representing moderate lag-1 correlation.

For the main experiments, we fixed the number of sampled pairs per uncensored record at $s = 10$ and used mini-batches of size $b = 64$ for stochastic optimization of the Gehan-type loss. Model parameters were trained using the Adam optimizer (Kingma and Ba, 2014) with learning rate $\alpha = 10^{-4}$, and training was run for $2, 3, 5, 10$, or 15 epochs to assess the effect of training duration on predictive performance. For each scenario, we generated independent training and test datasets. Training sample sizes were $n_{\text{train}} = 1000$ and $n_{\text{train}} = 5000$, while the test set size was fixed at $n_{\text{test}} = 2000$. The number of gap times per subject was sampled from a Poisson distribution with mean $\lambda = 2$. Predictive performance was assessed using the adjusted mean squared error (AMSE) and the inverse-probability-of-censoring weighted (IPCW) concordance index. The classical C-index, introduced by Harrell et al. (Harrell et al., 1996), measures how well a prognostic score or biomarker predicts a censored survival outcome and is similar to the area under the receiver operating characteristic curve (AUC) (Hastie et al., 2009; Egan, 1975; Fawcett, 2006; Hanley and McNeil, 1982). Both AMSE and C-index have been commonly used in single-event nonlinear AFT models, including recent work (Norman et al., 2024; Kim et al., 2024).

The AMSE was computed as

$$\text{AMSE} = \frac{1}{\sum_{i=1}^{n_{\text{test}}} K_i^*} \sum_{i=1}^{n_{\text{test}}} \sum_{j=1}^{K_i^*} \frac{\delta_{ij}}{\widehat{KM}(S_{ij})} \left(Y_{ij} - \widehat{\mu}_{ij}\right)^2, \tag{3.5}$$

where $\widehat{\mu}_{ij}$ is the model prediction for Y_{ij}. The Kaplan–Meier estimate $\widehat{KM}(t)$ of the end of monitoring survival function corrects for incomplete observations. Observed events ($\delta_{ij} = 1$)

are upweighted by $1/\widehat{KM}(S_{ij})$ to reflect the reduced probability of being observed under censoring. Censored observations contribute nothing to the loss ($\delta_{ij} = 0$). In the absence of censoring, all weights equal one and the AMSE reduces to the standard MSE.

The IPCW C-index was defined as

$$\widehat{C}_{\text{IPCW}} = \frac{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \{\widehat{KM}(S_{ij})\}^{-2} \, \mathbb{I}\{G_{ij} < G_{kl}\} \, \mathbb{I}\{\widehat{\mu}_{ij} < \widehat{\mu}_{kl}\}}{\sum_{i \neq k} \sum_{j=1}^{K_i^*} \sum_{l=1}^{K_k^*} \delta_{ij} \{\widehat{KM}(S_{ij})\}^{-2} \mathbb{I}\{G_{ij} < G_{kl}\}}. \tag{3.6}$$

Intuitively, (3.6) compares all cross-subject pairs of retained gap-time records and counts a pair as concordant if the record with the smaller observed gap time also receives a smaller predicted log-gap-time mean. Inverse-probability-of-censoring weighting accounts for the fact that only a subset of observation pairs is fully comparable due to incomplete gap times at the end of the monitoring period. For right-censored survival data, the univariate version of the IPCW C-index is consistent under noninformative censoring (Uno et al., 2011; Liu et al., 2012). Following this, we use it to assess predictive discrimination for recurrent gap time data.

3.2 Results

3.2.1 Mean Function Estimation

The results across Tables 1–3 illustrate the robust performance of the RNN-AGT model under diverse data-generating mechanisms, error distributions, censoring levels, and dependence structures. For the nonlinear mean function with interactions from Table 1, the model consistently achieves high IPCW C-indices (typically 0.85–0.94), indicating strong ranking ability even under heavy censoring and within-subject dependence. As expected, AMSE increases with censoring from 25% to 65%, reflecting reduced effective sample size after IPCW adjustment, while the C-index remains relatively stable, highlighting robustness in maintaining relative risk ordering. Performance improves with additional epochs but eventually plateaus or slightly declines under 50–65% censoring for smaller training sets ($n_{\text{train}} = 1000$), indicating mild overfitting; in contrast, larger training sets ($n_{\text{train}} = 5000$) achieve near-optimal performance in just 2–5 epochs, underscoring the RNN's data efficiency and fast convergence. Across error distributions, Gumbel errors generally result in lower AMSE and comparable or higher C-indices, while Logistic errors are more challenging, particularly under heavy censoring. Furthermore, AR(1) dependence consistently yields slightly higher C-indices than frailty-based correlation, suggesting that the RNN effectively captures serial dependence beyond subject-level random effects.

Table 2 presents a more challenging scenario, where the true mean function exhibits smooth nonlinearities in a GAM-type form. Here, the model shows a moderate decline in both C-index and AMSE across all configurations, reflecting increased complexity and the difficulty of learning smooth nonlinear effects. For smaller training sets ($n_{\text{train}} = 1000$), C-indices range from approximately 0.72 to 0.86 under Normal and Gumbel errors, and decrease further under Logistic errors as censoring increases, while AMSE rises steadily. Longer training epochs do not consistently improve performance; in several cases, particularly with 50–65% censoring, extended training leads to declining C-indices despite modest

AMSE reductions, indicating overfitting with limited effective sample size. Increasing the training sample size ($n_{\text{train}} = 5000$) stabilizes performance, but improvements plateau beyond 3–5 epochs, emphasizing the importance of early stopping. As in the interaction-based setting, Gumbel errors produce lower AMSE, while Logistic errors remain the most challenging. AR(1) dependence again slightly outperforms frailty-based correlation, demonstrating effective exploitation of temporal dependence.

Finally, Table 3 corresponds to the most favorable scenario, where the true mean function is linear and well aligned with the AFT structure. In this case, the RNN-AGT model attains uniformly high C-indices, typically above 0.90 and frequently exceeding 0.95 under Normal and Gumbel errors, even with 65% censoring. Compared to nonlinear and GAM-type settings, performance is more stable across training epochs, sample sizes, and censoring levels, demonstrating the model's ability to recover linear effects efficiently. While AMSE still increases with heavier censoring, this primarily reflects IPCW variability rather than deterioration in ranking performance. Longer training epochs improve C-index modestly at early stages but lead to substantial AMSE inflation at later epochs, particularly for $n_{\text{train}} = 5000$, highlighting overfitting in the squared-error scale without affecting concordance. Error distribution effects are minimal, with Logistic errors slightly more challenging but still producing strong C-indices above 0.89. AR(1) dependence continues to confer a modest advantage over frailty-based correlation.

Overall, these results collectively demonstrate the flexibility, robustness, and predictive strength of the RNN-AGT model. It maintains high discriminative performance across linear, interaction-based, and nonlinear GAM-type mean structures, adapts efficiently to varying sample sizes, and is resilient to heavy censoring and complex within-subject dependence. The model is particularly effective in capturing temporal correlations via its recurrent architecture and shows robustness across different error distributions, with Gumbel errors often yielding the most favorable AMSE. In summary, the simulation study confirms that the RNN-AGT framework provides a powerful and generalizable approach for predicting recurrent event times under diverse and realistic scenarios.

3.2.2 Loss Behavior Across Epochs

To examine the learning dynamics of the RNN-AGT model, we study the evolution of the training loss across epochs under varying data-generating mechanisms. We fix the training sample size at $n_{\text{train}} = 1000$ and the censoring proportion at 25%. For each configuration, the model is trained for 30 epochs, and the training loss is recorded under three error distributions: Normal, Gumbel, and Logistic.

For each mean function—Nonlinear with interactions, GAM-type Nonlinear, and Linear—the results are displayed using a common layout in which rows correspond to error distributions. This organization facilitates direct comparison of convergence behavior across distributions within each modeling scenario.

Across all three mean functions, the training loss exhibits a stable and generally monotone decrease over epochs, indicating well-behaved optimization and effective learning.

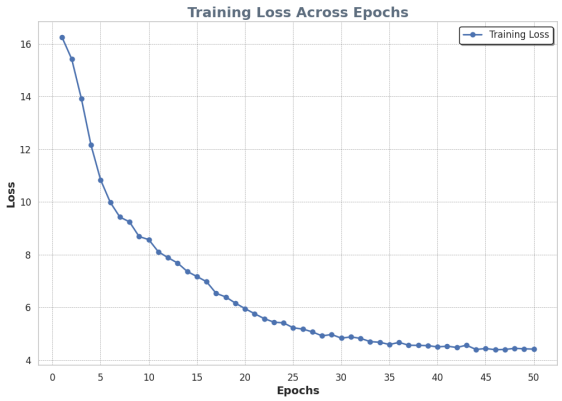
Comparing mean structures, the Linear mean function converges most rapidly and attains the lowest loss levels, reflecting its relative simplicity. The GAM-type Nonlinear mean function shows slightly slower convergence, while the Nonlinear + Interaction mean func-

Table 1: Simulation results (IPCW C-index / AMSE) for the nonlinear mean function with interaction under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.849/7.58	0.881/6.52	0.909/5.81	0.879/6.42	0.903/6.35	0.931/7.13
		50	0.863/10.91	0.863/10.64	0.860/13.75	0.855/11.13	0.863/13.40	0.878/18.49
		65	0.891/23.02	0.895/19.80	0.878/18.85	0.899/17.69	0.880/19.56	0.871/23.84
	AR(1)	25	0.851/7.37	0.892/5.92	0.918/6.39	0.882/6.69	0.904/6.17	0.941/6.77
		50	0.874/13.39	0.877/11.86	0.886/13.73	0.864/11.96	0.877/13.09	0.903/18.00
		65	0.901/21.68	0.897/18.10	0.883/18.29	0.903/17.78	0.888/19.27	0.880/24.57
Gumbel	Frailty	25	0.852/8.56	0.887/5.84	0.905/4.77	0.884/6.22	0.905/5.14	0.930/4.14
		50	0.880/11.04	0.892/9.14	0.882/10.83	0.879/9.36	0.865/11.19	0.865/13.97
		65	0.904/20.57	0.902/14.83	0.907/15.27	0.900/14.58	0.900/15.25	0.859/19.72
	AR(1)	25	0.843/8.72	0.895/4.98	0.934/3.08	0.893/5.32	0.912/4.47	0.940/3.56
		50	0.884/9.80	0.899/8.07	0.902/8.24	0.884/8.202	0.887/10.07	0.906/12.42
		65	0.904/19.00	0.919/15.23	0.902/13.39	0.907/13.69	0.906/13.47	0.881/18.58
Logistic	Frailty	25	0.848/10.18	0.875/8.63	0.891/7.62	0.870/9.40	0.887/ 8.74	0.904/8.35
		50	0.878/15.36	0.874/13.42	0.845/15.72	0.863/13.58	0.838/16.08	0.817/19.43
		65	0.893/26.39	0.887/20.94	0.867/21.06	0.870/21.23	0.858/22.36	0.776/27.91
	AR(1)	25	0.933/9.24	0.884/8.04	0.898/7.50	0.882/8.60	0.898/8.39	0.920/8.14
		50	0.886/14.75	0.889/13.03	0.873/15.71	0.878/13.26	0.873/15.70	0.873/18.95
		65	0.898/25.38	0.912/21.07	0.900/20.36	0.893/20.69	0.886/21.50	0.841/27.30

Table 2: Simulation results (IPCW C-index / AMSE) for the GAM-type Nonlinear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.841/8.30	0.837/6.57	0.830/5.83	0.836/6.68	0.830/6.80	0.831/10.63
		50	0.827/11.58	0.770/10.28	0.722/9.15	0.794/9.99	0.732/9.61	0.685/12.64
		65	0.849/16.22	0.768/14.16	0.698/13.59	0.806/15.97	0.740/14.40	0.595/13.97
	AR(1)	25	0.856/7.89	0.860/6.08	0.861/5.41	0.860/6.30	0.859/6.408	0.859/9.29
		50	0.853/11.28	0.828/9.46	0.817/8.33	0.841/9.79	0.805/9.54	0.754/11.63
		65	0.874/16.73	0.825/14.72	0.785/13.48	0.855/16.25	0.810/15.59	0.690/14.04
Gumbel	Frailty	25	0.824/7.63	0.816/5.19	0.815/3.89	0.814/4.66	0.804/4.11	0.806/6.71
		50	0.789/8.47	0.735/6.67	0.697/5.40	0.751/6.30	0.689/5.32	0.629/6.48
		65	0.809/10.73	0.740/8.98	0.704/7.73	0.760/9.28	0.714/8.15	0.612/7.24
	AR(1)	25	0.844/7.16	0.848/4.89	0.850/3.55	0.850/4.45	0.847/3.74	0.851/5.57
		50	0.850/8.10	0.827/6.03	0.795/4.75	0.814/6.03	0.776/5.00	0.740/5.89
		65	0.872/9.842	0.810/8.60	0.792/7.17	0.826/8.71	0.797/7.40	0.666/7.43
Logistic	Frailty	25	0.787/7.62	0.774/6.34	0.752/5.92	0.746/6.14	0.726/6.14	0.704/9.66
		50	0.756/9.17	0.686/7.92	0.655/7.45	0.667/8.22	0.619/8.13	0.538/10.44
		65	0.781/11.74	0.689/10.53	0.632/10.09	0.690/11.77	0.607/11.10	0.526/11.83
	AR(1)	25	0.80/6.84	0.812/5.64	0.809/5.35	0.804/5.71	0.791/5.99	0.783/8.86
		50	0.762/7.40	0.756/7.32	0.724/6.93	0.764/7.53	0.699/7.27	0.617/9.20
		65	0.843/11.15	0.769/9.80	0.719/9.91	0.778/10.61	0.729/10.48	0.584/11.25



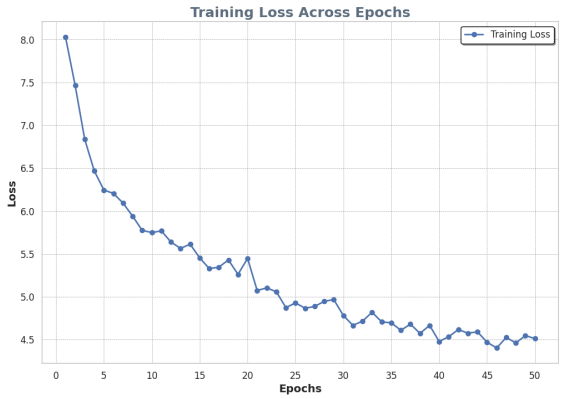
(a) Normal errors

(b) Gumbel errors

(c) Logistic errors

Figure 1: Training loss across epochs for the Nonlinear + Interaction mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.

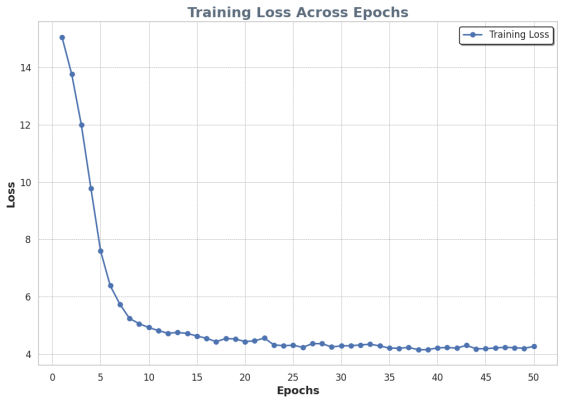
(a) Normal errors



(b) Gumbel errors

(c) Logistic errors

Figure 2: Training loss across epochs for the GAM-type Nonlinear mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.



(a) Normal errors

(b) Gumbel errors

(c) Logistic errors

Figure 3: Training loss across epochs for the Linear mean function under different error distributions, with $n_{\text{train}} = 1000$ and 25% censoring.

Table 3: Simulation results (IPCW C-index / AMSE) for the Linear mean function under different error distributions, censoring levels, and correlation structures.

Error Dist.	Structure	$\%C$	$n_{\text{train}} = 1000$			$n_{\text{train}} = 5000$		
			Epoch 5	Epoch 10	Epoch 15	Epoch 2	Epoch 3	Epoch 5
Normal	Frailty	25	0.924/10.51	0.941/14.93	0.944/18.30	0.942/15.64	0.944/19.42	0.943/27.87
		50	0.941/22.94	0.924/25.56	0.897/32.49	0.940/25.78	0.908/34.07	0.902/46.16
		65	0.940/40.89	0.957/34.68	0.934/38.37	0.961/34.23	0.942/39.57	0.877/53.82
	AR(1)	25	0.924/10.79	0.948/14.75	0.95/18.20	0.949/15.18	0.951/19.77	0.951/27.76
		50	0.941/22.91	0.935/26.09	0.924/31.56	0.945/25.09	0.929/32.62	0.927/43.53
		65	0.941/40.26	0.961/33.57	0.942/38.19	0.962/34.07	0.944/39.41	0.902/53.12
Gumbel	Frailty	25	0.917/8.80	0.926/11.85	0.928/16.43	0.933/11.46	0.936/15.16	0.930/22.56
		50	0.935/19.14	0.928/20.13	0.897/28.28	0.931/ 19.15	0.898/27.02	0.885/38.04
		65	0.942/35.76	0.958/28.39	0.940/32.53	0.953/28.73	0.939/32.51	0.858/47.92
	AR(1)	25	0.919/6.23	0.940/7.24	0.940/9.35	0.942/10.86	0.945/14.80	0.945/21.70
		50	0.940/18.93	0.941/19.70	0.919/27.84	0.940/20.25	0.922/26.70	0.907/38.51
		65	0.945/34.85	0.960/28.73	0.943/31.30	0.957/29.05	0.947/31.89	0.886/46.37
Logistic	Frailty	25	0.896/13.33	0.903/17.05	0.889/22.07	0.896/16.71	0.899/21.24	0.892/28.47
		50	0.926/28.12	0.880/30.09	0.825/37.16	0.891/30.09	0.848/37.23	0.811/48.57
		65	0.933/45.16	0.946/38.64	0.890/44.48	0.940/38.61	0.911/42.29	0.793/58.09
	AR(1)	25	0.899/13.94	0.914/18.47	0.919/21.74	0.916/17.79	0.918/22.82	0.916/28.80
		50	0.922/27.75	0.903/29.87	0.882/35.30	0.914/28.75	0.876/37.86	0.855/48.11
		65	0.933/44.19	0.947/39.21	0.899/45.48	0.944/38.13	0.925/43.67	0.836/58.00

tion exhibits higher initial loss and slower early-stage convergence, consistent with its increased structural complexity. Differences across error distributions are modest overall, though heavier-tailed distributions such as Gumbel and Logistic tend to produce higher initial losses and marginally slower convergence, particularly in the nonlinear settings.

Overall, these results demonstrate that the RNN-AGT model trains robustly across a range of mean functions and error distributions, with no evidence of instability or divergence for large sample sizes.

3.2.3 Model Performance Across Sub-sampling Settings

To examine the sensitivity of the RNN-AGT model to sub-sampling design choices, we conducted a simulation study varying both the number of sampled comparable pairs per uncensored residual, denoted by s, and the mini-batch size b. Specifically, we considered $s \in \{5, 10, 15\}$ and $b \in \{32, 64, 128\}$, while keeping all other modeling components fixed, including a nonlinear mean function with interactions, a normal error distribution, a single hidden layer, $\sigma_b = 0.5$, and a censoring rate of 25%. The learning rate was set to $\alpha = 10^{-4}$, and model performance was evaluated after 2, 3, 5, and 10 training epochs.

The objective of this analysis was to quantify the impact of sub-sampling parameters on both predictive accuracy and training stability. Model performance was assessed on an independent test set using the IPCW C-index to evaluate discriminative ability and the AMSE to assess calibration.

Table 4 reports the results for each combination of (s, b), with entries presented in the

Table 4: Sensitivity analysis of sub-sampling configurations. Test performance is reported as IPCW C-index / AMSE for different numbers of sampled pairs s, mini-batch sizes b, training sample sizes, and training epochs.

s	b	$n_{\text{train}} = 1000$		$n_{\text{train}} = 5000$	
		Epoch 5	Epoch 10	Epoch 2	Epoch 3
5	32	0.851/9.50	0.880/7.43	0.880/6.45	0.895/6.65
	64	0.836/14.69	0.856/8.93	0.848/7.68	0.873/6.61
	128	0.821/19.08	0.840/14.39	0.830/12.69	0.843/8.92
10	32	0.886/7.57	0.917/7.208	0.9123/6.10	0.934/6.01
	64	0.856/8.77	0.894/7.38	0.886/6.51	0.910/6.02
	128	0.840/14.20	0.864/8.43	0.857/7.35	0.881/6.56
15	32	0.901/7.67	0.938/6.48	0.934/6.28	0.941/8.80
	64	0.882/7.67	0.914/7.61	0.908/6.25	0.933/6.18
	128	0.850/10.24	0.886/7.61	0.883/6.58	0.904/6.27

form IPCW C-index / AMSE across different numbers of training epochs. This comparison highlights how increasing the number of sampled pairs per residual generally improves discrimination at the cost of higher computational burden, while larger mini-batch sizes tend to stabilize training but may slow convergence.

3.2.4 High-Dimensional Covariates Analysis

We evaluated the performance of the RNN-AGT model in high-dimensional settings, where the number of covariates p exceeds the number of signal features. Only the first five covariates contributed to the true mean function

$$f(\mathbf{z}_i) = 2z_{i1} - z_{i2} + 0.5z_{i3},$$

while the remaining covariates acted as pure noise. Covariates were generated as

$$z_{i1} \sim \text{Bernoulli}(0.5), \quad z_{ij} \sim \mathcal{N}\Big(z_{i,j-1}/3, 1\Big), \quad j = 2, \ldots, p,$$

introducing moderate correlation among features.

The training sample size was fixed at $n_{\text{train}} = 1000$, with $K_i \sim \text{Poisson}(\lambda = 2)$ gap times per subject. The test set contained $n_{\text{test}} = 2000$, Gaussian errors were used, and the censoring proportion was set to 25%. To assess robustness, we gradually increased the number of noise covariates from 0 to 100, generating the mean function as

$$f(\mathbf{z}) = z_1 - 6z_2 + 4z_3 + \sum_{k=4}^{K} \beta_k z_k, \quad \beta_k = 0,$$

so that the signal-to-noise ratio decreases with increasing p. Conventional AFT models are not expected to leverage irrelevant covariates, while deep neural networks can identify signal in high-dimensional spaces.

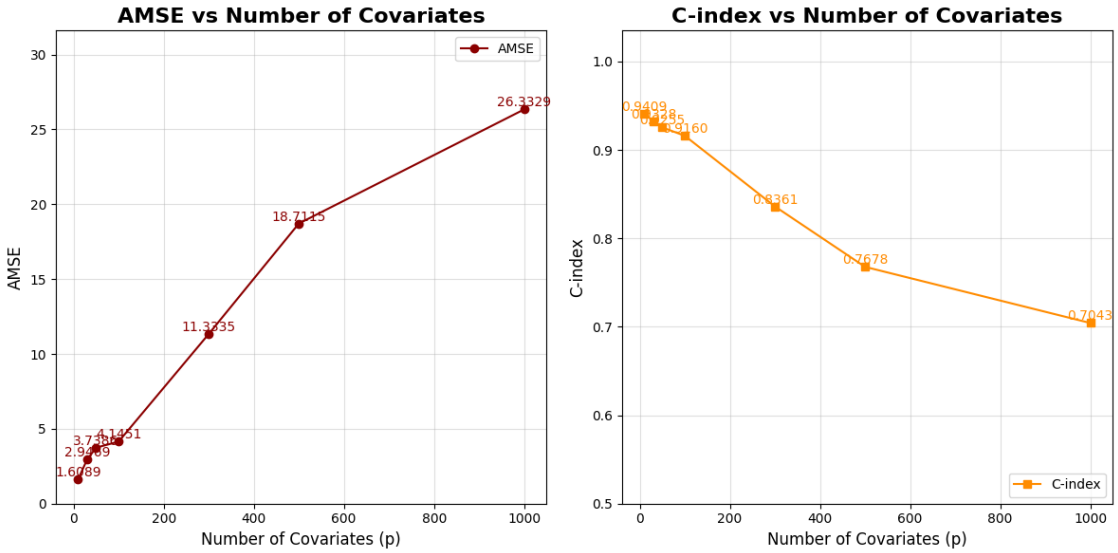


Figure 4: AMSE (left) and IPCW C-index (right) for the RNN-AGT model with increasing numbers of noise covariates.

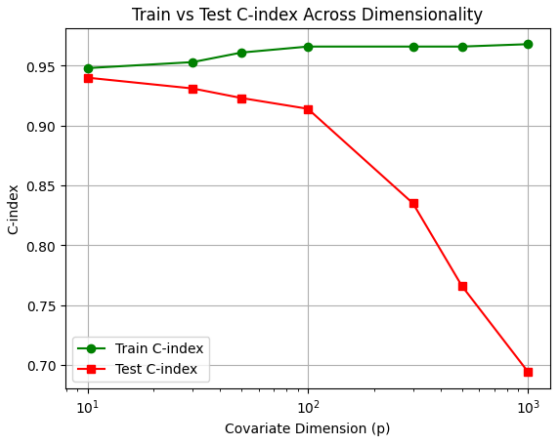
Figures 4–5c collectively illustrate RNN-AGT performance under high-dimensional scenarios. While the final AMSE and C-index remain strong for moderate p, test performance degrades for very high p, as evidenced by the train-test gaps (Figures 5a and 5b). The loss trajectories (Figure 5c) indicate that training remains stable and convergent, demonstrating that observed performance degradation is driven by statistical generalization limits rather than optimization failure. Overall, the RNN-AGT model is robust to high-dimensional noise covariates, maintaining accurate prediction and ranking where conventional models may fail.

4 Real Data Analysis

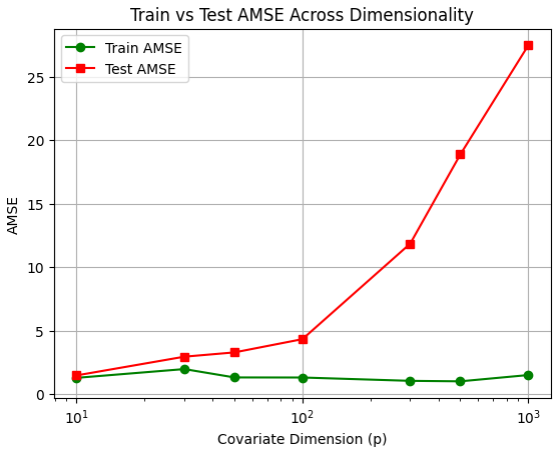
4.1 The Datasets

We evaluated the proposed RNN-AGT model on two real-world recurrent event datasets to demonstrate its ability to capture temporal dependence and patient heterogeneity.

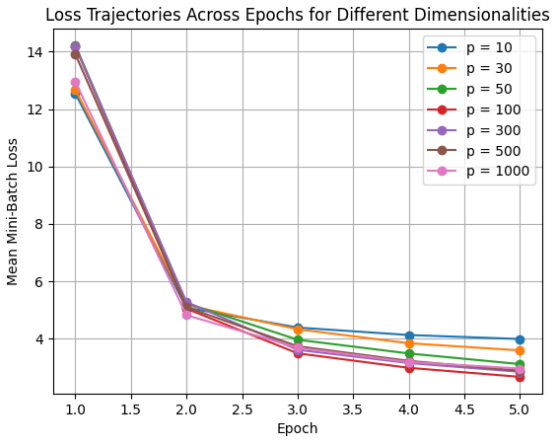
The first dataset examines repeated infections among **128** patients with Chronic Granulomatous Disease (CDG), publicly available from the `survival` R package (Therneau, 2020). Patients were followed from randomization until study end or dropout. The outcome of interest is the time to serious CGD infections, with each patient experiencing 0 to 7 events. Time intervals are recorded using `tstart` and `tstop`, with `status` indicating occurrence of infection, and additional covariates include `id`, `sex`, `age`, and `treat` (placebo vs. gamma interferon). The median follow-up time was 293 days, and the primary aim is to assess the effect of gamma interferon on the incidence of repeated infections.



(a) Train vs Test C-index



(b) Train vs Test AMSE



(c) Loss trajectories across epochs

Figure 5: (a) Train vs Test C-index, (b) Train vs Test AMSE, and (c) Loss trajectories across epochs for different covariate dimensions. The train-test gaps increase for high p, while loss convergence remains stable across all dimensions.

The second dataset consists of **403** colorectal cancer (CRC) patients following surgical treatment, originating from Gonzalez et al. (2005) and available through the `frailtypack` R package. Hospital readmissions were recorded as recurrent events, with right censoring occurring when patients were lost to follow-up or did not experience additional readmissions before study termination. Three time-fixed covariates were included: chemotherapy status (binary, 1 = received chemotherapy, 0 = no chemotherapy), Dukes' stage (1 = A–B, 2 = C, 3 = D), and sex (0 = male, 1 = female).

4.2 Predictive Performance

For both datasets, the RNN-AGT model was implemented using a gated recurrent unit (GRU) architecture with two recurrent layers and a hidden dimension of 64. Right censoring was handled via the pairwise mini-batch loss function with parameters $s = 30$ and $b = 64$. Model optimization used the RMSprop optimizer with a learning rate of 3×10^{-4}, and training was performed over 10 epochs on a CPU. Data were split into training (70%) and testing (30%) sets using a fixed random seed. Predictive accuracy was evaluated with the AMSE, and discriminative ability was assessed with the IPCW C-index, which accounts for censored observations.

Table 5: Predictive performance of the RNN-AGT model on two biomedical recurrent event datasets, evaluated using IPCW C-index and AMSE metrics.

Dataset	Train		Test	
	C-index	AMSE	C-index	AMSE
CGD Study (128 pts)	0.525	41.24	0.631	28.31
CRC Study (403 pts)	0.618	63.31	0.637	42.95

The results indicate that the RNN-AGT model achieves reasonable pointwise prediction (AMSE) and moderate discriminative ability (IPCW C-index) despite the small to moderate sample sizes and presence of censoring. The use of identical optimizer settings for both datasets demonstrates the model's robustness and stable generalization, effectively capturing recurrent event dynamics and covariate effects for clinical risk stratification and dynamic outcome prediction.

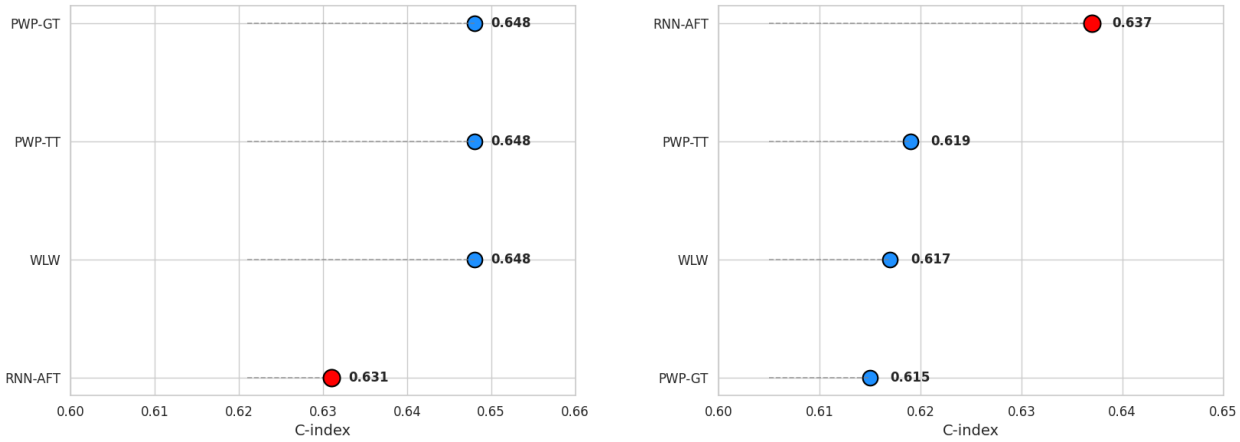
4.3 Benchmarking Against Cox-based Models

To benchmark the RNN-AGT model against classical recurrent event approaches, we compared its predictive discrimination to gap-time–based Prentice–Williams–Peterson (PWP) models and the Wei–Lin–Weissfeld (WLW) model using the C-index for both datasets.

For the CGD dataset with 128 patients, the RNN-AGT achieved a testing C-index of 0.631. In comparison, classical Cox-based models performed slightly better than random baseline but were very close to each other: WLW (0.648), PWP-TT (0.648), and PWP-GT (0.648). While these models handle recurrent events differently—WLW models marginal hazards for each event order, PWP-TT focuses on total times conditional on previous events, and

PWP-GT examines successive gap times—they all rely on a Cox-type multiplicative structure with a linear predictor, limiting their ability to capture nonlinear or history-dependent effects. The RNN-AGT effectively models complex temporal dependencies and nonlinear relationships, achieving competitive performance.

For the CRC dataset with 403 patients, the RNN-AGT achieved a testing C-index of 0.637, outperforming PWP-GT (0.615), PWP-TT (0.619), and WLW (0.617). Similar to the CGD study, classical Cox-based approaches often struggle to fully leverage nonlinear and patient-history information, whereas the RNN-AGT captures these effects effectively.



(a) CGD Dataset

(b) CRC Dataset

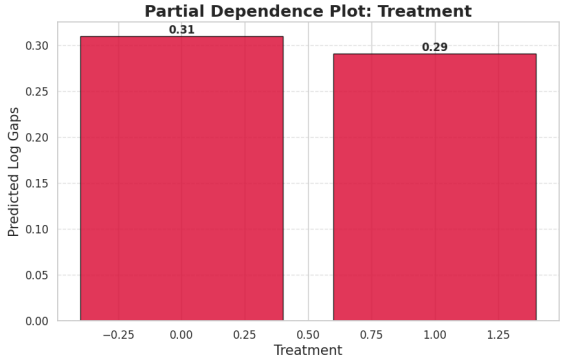
Figure 6: Predictive performance comparison of RNN-AGT, PWP-GT, PWP-TT, and WLW models measured by IPCW C-index across two biomedical recurrent event datasets.

4.4 Sensitivity Analysis

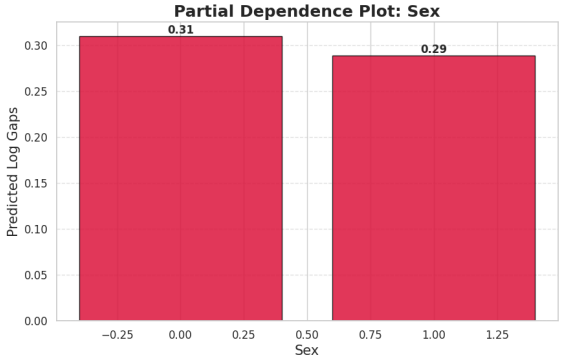
To enhance interpretability of the proposed RNN-AGT model, we examine partial dependence plots (PDPs) for key time-fixed covariates in both datasets. PDP analysis isolates the marginal effect of each covariate on the predicted log gap time while averaging over the empirical distribution of the remaining variables, providing insight into how individual patient characteristics influence recurrent event dynamics within the learned nonlinear model.

For the CGD study, patients receiving gamma interferon exhibit predicted gap times between infections that are comparable to those receiving placebo, suggesting no clear protective effect of treatment in this setting. Sex does not appear to have a meaningful influence on predicted infection intervals. In contrast, age shows a pronounced nonlinear effect: predicted gap times increase rapidly up to approximately 20 years of age and then gradually decrease thereafter, indicating a higher infection risk among older patients.

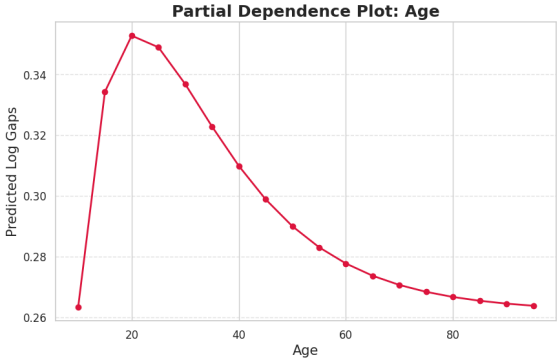
For the CRC study, patients receiving chemotherapy tend to have shorter predicted log gap times, suggesting a higher short-term risk of hospital readmission. Female patients exhibit longer predicted gap times than males, indicating a protective effect of sex. Higher Dukes' stages are associated with shorter predicted gaps, reflecting more frequent readmissions for advanced disease.



(a) Treatment (Gamma Interferon)

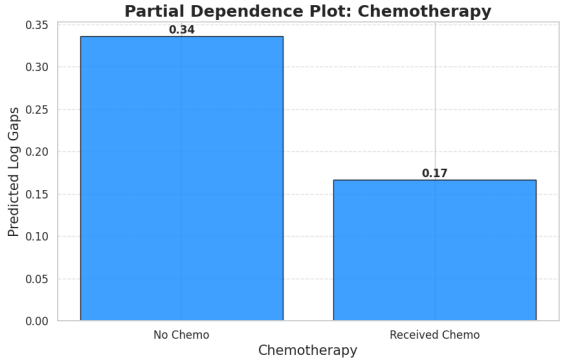


(b) Sex

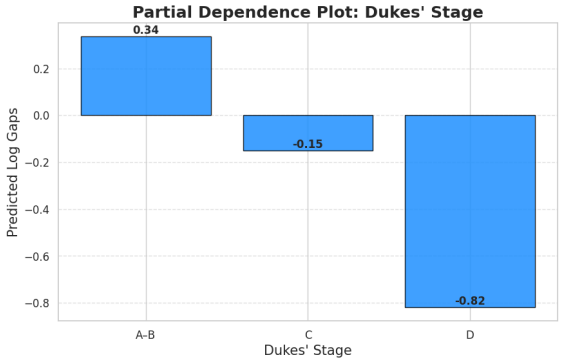


(c) Age at Randomization

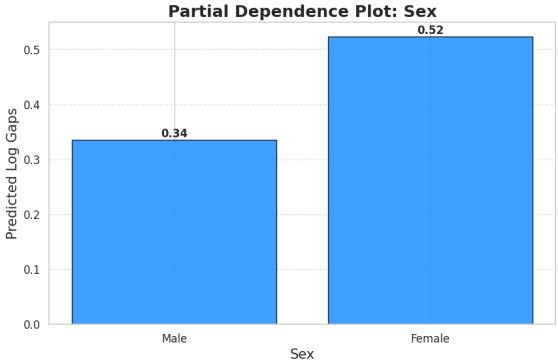
Figure 7: Partial dependence plots for the Chronic Granulomatous Disease dataset illustrating the marginal effect of treatment, sex, and age on predicted log gap times between infections.



(a) Chemotherapy status



(b) Dukes' stage



(c) Sex

Figure 8: Partial dependence plots for the Colorectal Cancer dataset showing the marginal effect chemotherapy status, Dukes' stage and sex on predicted log gap times between hospital readmissions.

These analyses demonstrate that the RNN-AGT model effectively captures nonlinear covariate effects and temporal dependencies in both datasets. The PDPs highlight clinically meaningful trends and support interpretability of model predictions for recurrent event outcomes.

5 Discussion

The RNN-AGT model demonstrates strong performance in modeling recurrent event data, effectively capturing nonlinear effects of time-fixed covariates while handling right-censoring and maintaining high concordance between predicted and observed event times. Our training approach balances pairwise loss optimization with absolute log-gap prediction, highlighting a trade-off between accurately ranking events and minimizing the average mean squared error (AMSE) of predictions. In practice, careful selection of the sub-sample size m for pairwise updates, coupled with early stopping criteria, can improve computational efficiency and prevent overfitting; however, larger sub-sample sizes increase both memory and runtime requirements. Despite these strengths, the current model does not explicitly accommodate time-dependent covariates or very long-range correlations between recurrent events, and excessive sub-sample sizes can exacerbate overfitting. Future work could explore adaptive or importance sampling strategies for pair selection and hybrid loss functions that combine rank-based and regression-based objectives to further improve prediction accuracy and robustness.

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